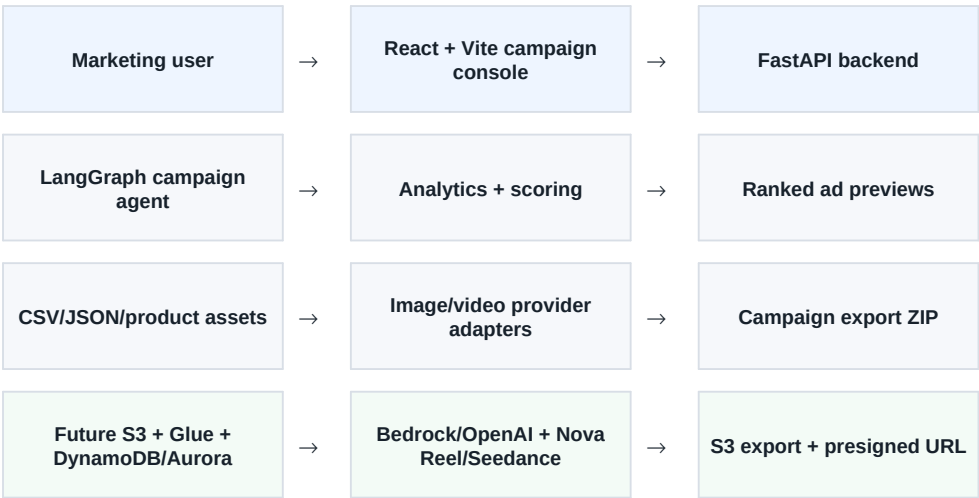


Architecture Overview

A one-page view of the local prototype and the AWS-ready migration path.



How Data Flows

- The frontend loads bootstrap data and renders the working campaign brief.
- FastAPI sends marketer actions to LangGraph workflows for analysis, recommendation, generation, and scoring.
- Optional AI steps refresh recommendations, refine the brief, polish copy, generate selected images, or create one selected video.
- The export endpoint packages manifest data, variants CSV, catalog references, and generated media into a local ZIP.

Key Runtime Services

- React/Vite UI for brief controls, ranked previews, rationale, media actions, and export.
- FastAPI backend for APIs, static files, provider adapters, and campaign packaging.
- LangGraph agent with load, analyze, recommend, generate, and score stages.
- Analytics layer for CTR, CVR, ROAS, CPA, segment fit, and forecasted lift.

AWS Migration Path

Prototype	AWS-ready target
Local CSV/JSON	S3, Glue, DynamoDB/Aurora
FastAPI process	App Runner, ECS, or Lambda container
React static build	S3 + CloudFront or Amplify
Local exports	S3 object + presigned URL
OpenAI/Seedance adapters	Secrets Manager + Bedrock/provider layer

Design Rationale

The prototype is intentionally portable and low-cost while preserving production boundaries: UI, API, agent orchestration, analytics, provider integrations, and export storage are separate. This lets the same workflow run locally now and move to AWS-managed data, media, model, and hosting services later.